

The Name of the Title Is Hope

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A clear and well-documented $\text{\LaTeX}$ document is presented as an article formatted for publication by ACM in a conference proceedings or journal publication. Based on the “acmart” document class, this article presents and explains many of the common variations, as well as many of the formatting elements an author may use in the preparation of the documentation of their work.

CCS Concepts: • **Do Not Use This Code** → **Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

ACM Reference Format:

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1 Introduction

ACM’s consolidated article template, introduced in 2017, provides a consistent $\text{\LaTeX}$ style for use across ACM publications, and incorporates accessibility and metadata-extraction functionality necessary for future Digital Library endeavors. Numerous ACM and SIG-specific $\text{\LaTeX}$ templates have been examined, and their unique features incorporated into this single new template.

If you are new to publishing with ACM, this document is a valuable guide to the process of preparing your work for publication. If you have published with ACM before, this document provides insight and instruction into more recent changes to the article template.

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The “acmart” document class can be used to prepare articles for any ACM publication — conference or journal, and for any stage of publication, from review to final “camera-ready” copy, to the author’s own version, with *very* few changes to the source.

2 Template Overview

As noted in the introduction, the “acmart” document class can be used to prepare many different kinds of documentation — a double-anonymous initial submission of a full-length technical paper, a two-page SIGGRAPH Emerging Technologies abstract, a “camera-ready” journal article, a SIGCHI Extended Abstract, and more — all by selecting the appropriate *template style* and *template parameters*.

This document will explain the major features of the document class. For further information, the *LaTeX User’s Guide* is available from <https://www.acm.org/publications/proceedings-template>.

2.1 Template Styles

The primary parameter given to the “acmart” document class is the *template style* which corresponds to the kind of publication or SIG publishing the work. This parameter is enclosed in square brackets and is a part of the `documentclass` command:

```
\documentclass[STYLE]{acmart}
```

Journals use one of three template styles. All but three ACM journals use the `acmsmall` template style:

- `acmsmall`: The default journal template style.
- `acmlarge`: Used by JOCCH and TAP.
- `acmtog`: Used by TOG.

The majority of conference proceedings documentation will use the `acmconf` template style.

- `sigconf`: The default proceedings template style.
- `sigchi`: Used for SIGCHI conference articles.
- `sigplan`: Used for SIGPLAN conference articles.

2.2 Template Parameters

In addition to specifying the *template style* to be used in formatting your work, there are a number of *template parameters* which modify some part of the applied template style. A complete list of these parameters can be found in the *LaTeX User’s Guide*.

Frequently-used parameters, or combinations of parameters, include:

- `anonymous, review`: Suitable for a “double-anonymous” conference submission. Anonymizes the work and includes line numbers. Use with the `\acmSubmissionID` command to print the submission’s unique ID on each page of the work.
- `authorversion`: Produces a version of the work suitable for posting by the author.
- `screen`: Produces colored hyperlinks.

This document uses the following string as the first command in the source file:

```
\documentclass[acmsmall,natbib=false]{acmart}
```

3 Modifications

Modifying the template — including but not limited to: adjusting margins, typeface sizes, line spacing, paragraph and list definitions, and the use of the `\vspace` command to manually adjust the vertical spacing between elements of your work — is not allowed.

Your document will be returned to you for revision if modifications are discovered.

4 Typefaces

The “acmart” document class requires the use of the “Libertine” typeface family. Your $\text{\TeX}$ installation should include this set of packages. Please do not substitute other typefaces. The “lmodern” and “ltimes” packages should not be used, as they will override the built-in typeface families.

5 Title Information

The title of your work should use capital letters appropriately - <https://capitalizemytitle.com/> has useful rules for capitalization. Use the `title` command to define the title of your work. If your work has a subtitle, define it with the `subtitle` command. Do not insert line breaks in your title.

If your title is lengthy, you must define a short version to be used in the page headers, to prevent overlapping text. The `title` command has a “short title” parameter:

```
\title[short title]{full title}
```

6 Authors and Affiliations

Each author must be defined separately for accurate metadata identification. As an exception, multiple authors may share one affiliation. Authors’ names should not be abbreviated; use full first names wherever possible. Include authors’ e-mail addresses whenever possible.

Grouping authors’ names or e-mail addresses, or providing an “e-mail alias,” as shown below, is not acceptable:

```
\author{Brooke Aster, David Mehldau}
\email{dave,judy,steve@university.edu}
\email{firstname.lastname@phillips.org}
```

The `authornote` and `authornotemark` commands allow a note to apply to multiple authors — for example, if the first two authors of an article contributed equally to the work.

If your author list is lengthy, you must define a shortened version of the list of authors to be used in the page headers, to prevent overlapping text. The following command should be placed just after the last `\author{}` definition:

```
\renewcommand{\shortauthors}{McCartney, et al.}
```

Omitting this command will force the use of a concatenated list of all of the authors’ names, which may result in overlapping text in the page headers.

The article template’s documentation, available at <https://www.acm.org/publications/proceedings-template>, has a complete explanation of these commands and tips for their effective use.

Note that authors’ addresses are mandatory for journal articles.

7 Rights Information

Authors of any work published by ACM will need to complete a rights form. Depending on the kind of work, and the rights management choice made by the author, this may be copyright transfer, permission, license, or an OA (open access) agreement.

Regardless of the rights management choice, the author will receive a copy of the completed rights form once it has been submitted. This form contains $\text{\LaTeX}$ commands that must be copied into the source document. When the document source is compiled, these commands and their parameters add formatted text to several areas of the final document:

- the “ACM Reference Format” text on the first page.
- the “rights management” text on the first page.
- the conference information in the page header(s).

Rights information is unique to the work; if you are preparing several works for an event, make sure to use the correct set of commands with each of the works.

The ACM Reference Format text is required for all articles over one page in length, and is optional for one-page articles (abstracts).

8 CCS Concepts and User-Defined Keywords

Two elements of the “acmart” document class provide powerful taxonomic tools for you to help readers find your work in an online search.

The ACM Computing Classification System — <https://www.acm.org/publications/class-2012> — is a set of classifiers and concepts that describe the computing discipline. Authors can select entries from this classification system, via <https://dl.acm.org/ccs/ccs.cfm>, and generate the commands to be included in the $\LaTeX$ source.

User-defined keywords are a comma-separated list of words and phrases of the authors’ choosing, providing a more flexible way of describing the research being presented.

CCS concepts and user-defined keywords are required for all articles over two pages in length, and are optional for one- and two-page articles (or abstracts).

9 Sectioning Commands

Your work should use standard $\LaTeX$ sectioning commands: `\section`, `\subsection`, `\subsubsection`, `\paragraph`, and `\subparagraph`. The sectioning levels up to `\subsubsection` should be numbered; do not remove the numbering from the commands.

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is **not allowed**.

Below are examples of sectioning commands.

9.1 Subsection

This is a subsection.

9.1.1 Subsubsection. This is a subsubsection.

Paragraph. This is a paragraph.

Subparagraph This is a subparagraph.

10 Tables

The “acmart” document class includes the “booktabs” package — <https://ctan.org/pkg/booktabs> — for preparing high-quality tables.

Table captions are placed *above* the table.

Because tables cannot be split across pages, the best placement for them is typically the top of the page nearest their initial cite. To ensure this proper “floating” placement of tables, use the environment **table** to enclose the table’s contents and the table caption. The contents of the table itself must go in the **tabular** environment, to be aligned properly in rows and columns, with the desired horizontal and vertical rules. Again, detailed instructions on **tabular** material are found in the *$\LaTeX$ User’s Guide*.

Immediately following this sentence is the point at which Table 1 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page’s live area, use the environment **table*** to enclose the table’s contents and the table caption. As with a single-column table, this wide table will “float” to a location deemed more desirable. Immediately following this sentence

Table 1. Frequency of Special Characters

Non-English or Math	Frequency	Comments
Ø	1 in 1,000	For Swedish names
π	1 in 5	Common in math
\$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

Table 2. Some Typical Commands

Command	A Number	Comments
<code>\author</code>	100	Author
<code>\table</code>	300	For tables
<code>\table*</code>	400	For wider tables

is the point at which Table 2 is included in the input file; again, it is instructive to compare the placement of the table here with the table in the printed output of this document.

Always use `midrule` to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognise table headers and support their users in navigating tables more easily.

11 Math Equations

You may want to display math equations in three distinct styles: inline, numbered or non-numbered display. Each of the three are discussed in the next sections.

11.1 Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the **math** environment, which can be invoked with the usual `\begin . . . \end` construction or with the short form `$ . . . $`. You can use any of the symbols and structures, from α to ω , available in $\text{\LaTeX}$ [Lamport 1986]; this section will simply show a few examples of in-text equations in context. Notice how this equation: $\lim_{n \rightarrow \infty} x = 0$, set here in in-line math style, looks slightly different when set in display style. (See next section).

11.2 Display Equations

A numbered display equation—one set off by vertical space from the text and centered horizontally—is produced by the **equation** environment. An unnumbered display equation is produced by the **displaymath** environment.

Again, in either environment, you can use any of the symbols and structures available in $\text{\LaTeX}$; this section will just give a couple of examples of display equations in context. First, consider the equation, shown as an inline equation above:

$$\lim_{n \rightarrow \infty} x = 0$$

(1)

Notice how it is formatted somewhat differently in the **displaymath** environment. Now, we'll enter an unnumbered equation:

$$\sum_{i=0}^{\infty} x + 1$$

and follow it with another numbered equation:

$$\sum_{i=0}^{\infty} x_i = \int_0^{\pi+2} f \quad (2)$$

just to demonstrate $\text{\LaTeX}$'s able handling of numbering.

12 Figures

The “figure” environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.



Fig. 1. 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (<https://goo.gl/VLCRBB>).

Your figures should contain a caption which describes the figure to the reader. Figure captions are placed *below* the figure.

Every figure should also have a figure description unless it is purely decorative. These descriptions convey what’s in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

A figure description must be unformatted plain text less than 2000 characters long (including spaces). **Figure descriptions should not repeat the figure caption – their purpose is to capture important information that is not already provided in the caption or the main text of the paper.** For figures that convey important and complex new information, a short text description may not be adequate. More complex alternative descriptions can be placed in an appendix and referenced in a short figure description. For example, provide a data table capturing the information in a bar chart, or a structured list representing a graph. For additional information regarding how best to write figure descriptions and why doing this is so important, please see <https://www.acm.org/publications/taps/describing-figures/>.

12.1 The “Teaser Figure”

A “teaser figure” is an image, or set of images in one figure, that are placed after all author and affiliation information, and before the body of the article, spanning the page. If you wish to have such a figure in your article, place the command immediately before the `\maketitle` command:

```
\begin{teaserfigure}
\includegraphics[width=\textwidth]{sampleteaser}
\caption{figure caption}
\Description{figure description}
\end{teaserfigure}
```

13 Citations and Bibliographies

The use of Bib $\TeX$ for the preparation and formatting of one’s references is strongly recommended. Authors’ names should be complete — use full first names (“Donald E. Knuth”) not initials (“D. E. Knuth”) — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

Using the Bib $\LaTeX$ system, the bibliography is included in your source document with the following command, placed just before the `\end{document}` command:

```
\printbibliography
```

The command `\addbibresource{bibfile}` declares the Bib $\TeX$ file to use in the **preamble** (before the command “`\begin{document}`”) of your $\LaTeX$ source where “bibfile” is the name, *with* the “.bib” suffix. Notice that `\addbibresource` takes only one argument: to declare multiple files, use multiple instances of the command.

Citations and references are numbered by default. A small number of ACM publications have citations and references formatted in the “author year” style; for these exceptions, please pass the option `style=acmauthoryear` to the `biblatex` package loaded in the **preamble** (before the command “`\begin{document}`”) of your $\LaTeX$ source.

Some examples. A paginated journal article [Abril and Plant 2007], an enumerated journal article [S. Cohen et al. 2007], a reference to an entire issue [J. Cohen 1996], a monograph (whole book) [Kosiur 2001], a monograph/whole book in a series (see 2a in spec. document) [Harel 1979], a divisible-book such as an anthology or compilation [Editor 2007] followed by the same example, however we only output the series if the volume number is given [Editor 2008] (so Editor00a’s series should NOT be present since it has no vol. no.), a chapter in a divisible book [Spector 1990], a chapter in a divisible book in a series [Douglass et al. 1998], a multi-volume work as book [Knuth 1997], a couple of articles in a proceedings (of a conference, symposium, workshop for example)

(paginated proceedings article) [Andler 1979; Hagerup et al. 1993], a proceedings article with all possible elements [Smith 2010], an example of an enumerated proceedings article [Gundy et al. 2007], an informally published work [Harel 1978], a couple of preprints [Anzaroot, Passos, et al. 2014; Bornmann et al. 2019], a doctoral dissertation [Clarkson 1985], a master’s thesis: [Anisi 2003], an online document / world wide web resource [Ablamowicz and Fauser 2007; Poker-Edge.Com 2006; Thornburg 2001], a video game (Case 1) [Obama 2008] and (Case 2) [Novak 2003] and [Lee 2005] and (Case 3) a patent [Scientist 2009], work accepted for publication [Rous 2008], ‘YYYYb’-test for prolific author [Saeedi, Zamani, and Sedighi 2010] and [Saeedi, Zamani, Sedighi, and Sasanian 2010]. Other cites might contain ‘duplicate’ DOI and URLs (some SIAM articles) [Kirschmer and Voight 2010]. Boris / Barbara Beeton: multi-volume works as books [Hörmander 1985b] and [Hörmander 1985a]. A presentation [Reiser 2014]. An article under review [Baggett et al. 2025]. A couple of citations with DOIs: [“IEEE TCSC Executive Committee” 2004; Kirschmer and Voight 2010]. Online citations: [Thornburg 2001; *Institutional members of the $T_{\text{E}}\text{X}$ Users Group* 2017; Veytsman 2017]. Data Artifacts: [Anzaroot and McCallum 2013]. Software project: [Delebecque et al. 1994; The CGAL Project 1996]. Software Version: [Greenman and Felleisen 2020]. Software Module: [Karavelas 2020]. Code fragment: [Di Cosmo and Danelutto 2020].

14 Acknowledgments

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document.

This section has a special environment:

```
\begin{acks}
...
\end{acks}
```

so that the information contained therein can be more easily collected during the article metadata extraction phase, and to ensure consistency in the spelling of the section heading.

Authors should not prepare this section as a numbered or unnumbered `\section`; please use the “acks” environment.

15 Appendices

If your work needs an appendix, add it before the “`\end{document}`” command at the conclusion of your source document.

Start the appendix with the “appendix” command:

```
\appendix
```

and note that in the appendix, sections are lettered, not numbered. This document has two appendices, demonstrating the section and subsection identification method.

16 Multi-language papers

Papers may be written in languages other than English or include titles, subtitles, keywords and abstracts in different languages (as a rule, a paper in a language other than English should include an English title and an English abstract). Use `language=...` for every language used in the paper. The last language indicated is the main language of the paper. For example, a French paper with additional titles and abstracts in English and German may start with the following command

```
\documentclass[sigconf, language=english, language=german,
language=french]{acmart}
```


The title, subtitle, keywords and abstract will be typeset in the main language of the paper. The commands `\translatedXXX`, `XXX` begin title, subtitle and keywords, can be used to set these elements in the other languages. The environment `translatedabstract` is used to set the translation of the abstract. These commands and environment have a mandatory first argument: the language of the second argument. See `sample-sigconf-i13n.tex` file for examples of their usage.

Acknowledgments

To Robert, for the bagels and explaining CMYK and color spaces.

Ethics and Privacy Statement

This section of your ACM work should discuss the potential societal risks that might result from its publication; two to three sentences related to the findings of your study, or new advancements made possible by their developed methods. The privacy and ethics statement should clearly address the broader impacts of their work as it relates to the authors' interpretation of privacy, fairness, safety, human rights, data sovereignty, or future misuse and any benefit/risk trade-off resulting from this research. We acknowledge that some papers may have minimal societal risks beyond those considered by institutional review boards, and the dimensions considered by any review of the user study design or dataset licenses could be provided in this statement.

References

- Rafal Ablamowicz and Bertfried Fauser. 2007. *CLIFFORD: a Maple 11 Package for Clifford Algebra Computations, version 11*. Retrieved February 28, 2008 from <http://math.tntech.edu/rafal/cliff11/index.html>.
- Patricia S. Abril and Robert Plant. Jan. 2007. "The patent holder's dilemma: Buy, sell, or troll?" *Communications of the ACM*, 50, 1, (Jan. 2007), 36–44. doi:10.1145/1188913.1188915.
- Sten Andler. 1979. "Predicate Path expressions." In: *Proceedings of the 6th. ACM SIGACT-SIGPLAN symposium on Principles of Programming Languages (POPL '79)*. ACM Press, New York, NY, 226–236. doi:10.1145/567752.567774.
- David A. Anisi. 2003. "Optimal Motion Control of a Ground Vehicle." Master's thesis. Royal Institute of Technology (KTH), Stockholm, Sweden.
- Sam Anzaroot and Andrew McCallum. 2013. *UMass Citation Field Extraction Dataset*. Retrieved May 27, 2019 from <http://www.iesl.cs.umass.edu/data/data-umasscitationfield>.
- Sam Anzaroot, Alexandre Passos, David Belanger, and Andrew McCallum. 2014. *Learning Soft Linear Constraints with Application to Citation Field Extraction*. (2014). arXiv: 1403.1349. doi:10.48550/arXiv.1403.1349.
- R. Baggett, M. Simecek, C. Chambellan, K. Tsui, and M. Fraune. 2025. *Fluidity in the Phased Framework of Technology Acceptance: Case Study to Gain a Holistic Understanding of (Older Adult) Participant Advancement Through Acceptance Phases with Mobile Telepresence Robots*. (2025).
- Lutz Bornmann, K. Brad Wray, and Robin Haunschild. May 2019. *Citation concept analysis (CCA)—A new form of citation analysis revealing the usefulness of concepts for other researchers illustrated by two exemplary case studies including classic books by Thomas S. Kuhn and Karl R. Popper*. (May 2019). arXiv: 1905.12410 [cs.DL].
- Kenneth L. Clarkson. 1985. "Algorithms for Closest-Point Problems (Computational Geometry)." Ph.D. Dissertation. Stanford University, Palo Alto, CA. UMI Order Number: AAT 8506171.
- Jacques Cohen, (Ed.). Nov. 1996. *Special issue: Digital Libraries. Commun. ACM* 39, 11 (Nov. 1996).
- Sarah Cohen, Werner Nutt, and Yehoshua Sagie. Apr. 2007. "Deciding equivalences among conjunctive aggregate queries." *J. ACM*, 54, 2, Article 5, (Apr. 2007), 50 pages. doi:10.1145/1219092.1219093.
- [SW] François Delebecque, Claude Gomez, Maurice Goursat, Ramine Nikoukhah, Serge Steer, and Jean-Philippe Chancelier, *Scilab* 1994. Inria. HAL: (hal-02090402v1), URL: <https://www.scilab.org/>, vcs: <https://github.com/scilab/scilab>.
- [SW exc.] Roberto Di Cosmo and Marco Danelutto, "Core mapping routine," *The Parmap library* version 1.1.1, 2020. Inria, University of Paris, and University of Pisa. URL: <https://rdicosmo.github.io/parmap/>, vcs: <https://github.com/rdicosmo/parmap>; vis it=swh:1:snp:2a6c348c53eb77d458f24c9cbcecaf92e3c45615;anchor=swh:1:rel:373e2604d96de4ab1d505190b654c4045db773;path=/src/parmap.ml;lines=192-228).
- Bruce P. Douglass, David Harel, and Mark B. Trakhtenbrot. 1998. "Statecharts in use: structured analysis and object-orientation." In: *Lectures on Embedded Systems*. Lecture Notes in Computer Science. Vol. 1494. Ed. by Grzegorz Rozenberg and Frits W. Vaandrager. Springer-Verlag, London, 368–394. doi:10.1007/3-540-65193-4_29.

- Ed. by Ian Editor. "The title of book one. The book subtitle." (1st. ed.). Vol. 9. The name of the series one. University of Chicago Press, Chicago. Chap. The title of the chapter, 127–238. doi:10.1007/3-540-09237-4.
- Ed. by Ian Editor. "The title of book two. The book subtitle." (2nd. ed.). The name of the series two. University of Chicago Press, Chicago. Chap. 100, 25–137. doi:10.1007/3-540-09237-4.
- [SW Rel.] Ben Greenman and Matthias Felleisen, *tag-sound* version 4cc09ca, 2020. vcs: <https://github.com/nuprl/tag-sound>, swuid: {swh:1:dir:cd0b0abee707e57cd699e2e2ebd075da8ebf1f7;origin=https://github.com/nuprl/tag-sound;visit=swh:1:snp:7967bc0abee8bf3bfff9252207a07b73538525a;anchor=swh:1:rev:4cc09ca228947a99c8f4ac45eebf76e96ee96e53}.
- Matthew Van Gundy, Davide Balzarotti, and Giovanni Vigna. 2007. "Catch me, if you can: Evading network signatures with web-based polymorphic worms." In: *Proceedings of the first USENIX workshop on Offensive Technologies* (WOOT '07) Article Paper 7. USENIX Association, Berkley, CA, 9 pages.
- Torben Hagerup, Kurt Mehlhorn, and J. Ian Munro. 1993. "Maintaining Discrete Probability Distributions Optimally." In: *Proceedings of the 20th International Colloquium on Automata, Languages and Programming* (Lecture Notes in Computer Science). Vol. 700. Springer-Verlag, Berlin, 253–264.
- David Harel. 1979. *First-Order Dynamic Logic*. Lecture Notes in Computer Science. Vol. 68. Springer-Verlag, New York, NY. doi:10.1007/3-540-09237-4.
- David Harel. 1978. *LOGICS of Programs: AXIOMATICS and DESCRIPTIVE POWER*. MIT Research Lab Technical Report TR-200. Massachusetts Institute of Technology, Cambridge, MA.
- Lars Hörmander. 1985a. *The analysis of linear partial differential operators. III*. Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]. Vol. 275. Pseudodifferential operators. Springer-Verlag, Berlin, Germany, viii+525. ISBN: 3-540-13828-5.
- Lars Hörmander. 1985b. *The analysis of linear partial differential operators. IV*. Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]. Vol. 275. Fourier integral operators. Springer-Verlag, Berlin, Germany, vii+352. ISBN: 3-540-13829-3.
- "IEEE TCSC Executive Committee." In: *Proceedings of the IEEE International Conference on Web Services* (ICWS '04). IEEE Computer Society, Washington, DC, USA, 21–22. ISBN: 0-7695-2167-3. doi:<http://dx.doi.org/10.1109/ICWS.2004.64>.
- [SW Mod.] Menelaos Karavelas, "2D Voronoi Diagram Adaptor," *The Computational Geometry Algorithms Library* version 5.0.2 (Coord.by CGAL Editorial Board), 2020. URL: <https://doc.cgal.org/5.0.2/Manual/packages.html#PkgVoronoiDiagram2>, swuid: {swh:1:rel:636541bbf6c77863908eae744610a3d91fa58855;origin=https://github.com/CGAL/cgal}.
- Markus Kirschmer and John Voight. Jan. 2010. "Algorithmic Enumeration of Ideal Classes for Quaternion Orders." *SIAM J. Comput.*, 39, 5, (Jan. 2010), 1714–1747. doi:<https://doi.org/10.1137/080734467>.
- Donald E. Knuth. 1997. *The Art of Computer Programming, Vol. 1: Fundamental Algorithms (3rd. ed.)* Addison Wesley Longman Publishing Co., Inc., Boston.
- David Kosiur. 2001. *Understanding Policy-Based Networking*. (2nd. ed.). Wiley, New York, NY.
- Leslie Lamport. 1986. *TEX: A Document Preparation System*. Addison-Wesley, Reading, MA.
- Newton Lee. Jan. 2005. "Interview with Bill Kinder: January 13, 2005." *Comput. Entertain.*, 3, 1, (Jan. 2005), 4. doi:10.1145/1057270.1057278.
- Dave Novak. Mar. 2003. "Solder man." In: *ACM SIGGRAPH 2003 Video Review on Animation theater Program: Part I - Vol. 145 (July 27–27, 2003)*. ACM Press, New York, NY, (Mar. 2003), 4. <http://video.google.com/videoplay?docid=6528042696351994555>. doi:10.945/woot07-S422.
- Barack Obama. Mar. 2008. *A more perfect union*. Video. (Mar. 2008). Retrieved March 21, 2008 from <http://video.google.com/videoplay?docid=6528042696351994555>.
- Poker-Edge.Com. Mar. 2006. *Stats and Analysis*. (Mar. 2006). Retrieved June 7, 2006 from <http://www.poker-edge.com/stats.php>.
- Brian J. Reiser. 2014. *Designing coherent storylines aligned with NGSS for the K-12 classroom*. (2014). <https://www.academia.edu/6884962/>.
- Bernard Rous. July 2008. "The Enabling of Digital Libraries." *Digital Libraries*, 12, 3, Article Article 5, (July 2008). To appear.
- Mehdi Saeedi, Morteza Saheb Zamani, and Mehdi Sedighi. Apr. 2010. "A library-based synthesis methodology for reversible logic." *Microelectron. J.*, 41, 4, (Apr. 2010), 185–194.
- Mehdi Saeedi, Morteza Saheb Zamani, Mehdi Sedighi, and Zahra Sasanian. Dec. 2010. "Synthesis of Reversible Circuit Using Cycle-Based Approach." *J. Emerg. Technol. Comput. Syst.*, 6, 4, (Dec. 2010), 12 pages.
- Joseph Scientist. Aug. 2009. *The fountain of youth*. Patent No. 12345, Filed July 1st., 2008, Issued Aug. 9th., 2009. (Aug. 2009).
- Stan W. Smith. 2010. "An experiment in bibliographic mark-up: Parsing metadata for XML export." In: *Proceedings of the 3rd. annual workshop on Librarians and Computers* (LAC '10). Ed. by Reginald N. Smythe and Alexander Noble. Vol. 3. Paparazzi Press, Milan Italy, 422–431.
- Asad Z. Spector. 1990. "Achieving application requirements." In: *Distributed Systems*. (2nd. ed.). Ed. by Sape Mullender. ACM Press, New York, NY, 19–33. doi:10.1145/90417.90738.

- [SW] The CGAL Project, *The Computational Geometry Algorithms Library* (Coord.by CGAL Editorial Board), 1996. URL: <https://cgal.org/>.
- Harry Thornburg. Mar. 2001. *Introduction to Bayesian Statistics*. (Mar. 2001). Retrieved March 2, 2005 from <http://ccrma.stanford.edu/~jos/bayes/bayes.html>.
- Institutional members of the T_EX Users Group*. Retrieved May 27, 2017 from <http://wwwtug.org/instrmem.html>.
- Boris Veytsman. 2017. *acmart—Class for typesetting publications of ACM*. Retrieved May 27, 2017 from <http://www.ctan.org/pkg/acmart>.

A Research Methods

A.1 Part One

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A.2 Part Two

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B Online Resources

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